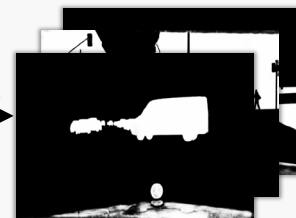


Input Image

VFM Backbone
 $f_{vfm}(\cdot)$

Decoder
 $g_{dec}(\cdot)$

Softmax



Prediction $P \in \mathcal{R}^{K \times H \times W}$

CE Loss

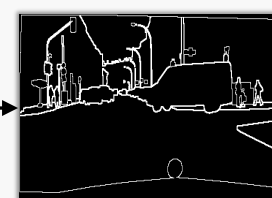
$\mathcal{L}_{ce}$

Local Structure

Loss $\mathcal{L}_{st}$



Label $Y \in \mathcal{R}^{H \times W}$



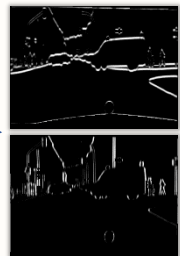
Boundary Region
 $B \in \mathcal{R}^{H \times W}$

Local Structure Regularization



Prediction P

Sobel
Gradient



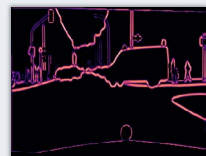
P_x

P_y

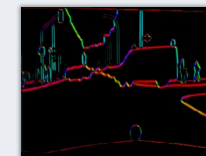
Structure Tensor

$$\begin{aligned} J_{xx} &= \sum_{k=0}^{K-1} P_{x,k}^2 \\ J_{yy} &= \sum_{k=0}^{K-1} P_{y,k}^2 \\ J_{xy} &= \sum_{k=0}^{K-1} P_{x,k} P_{y,k} \end{aligned}$$

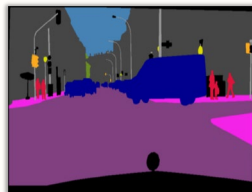
Local Structure of Prediction



Magnitude $\hat{E}_p$

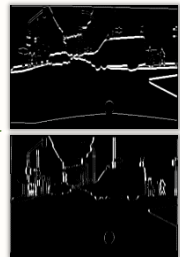


Orientation O_p



Label Y

One-hot &
Smoothing &
Sobel Gradient



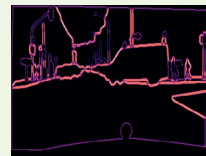
Y_x^{sm}

Y_y^{sm}

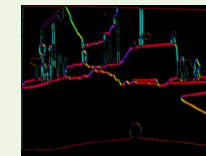
Structure Tensor

$$\begin{aligned} J_{xx} &= \sum_{k=0}^{K-1} Y_{x,k}^{sm} \\ J_{yy} &= \sum_{k=0}^{K-1} Y_{y,k}^{sm} \\ J_{xy} &= \sum_{k=0}^{K-1} Y_{x,k}^{sm} Y_{y,k}^{sm} \end{aligned}$$

Local Structure of Label



Magnitude $\hat{E}_y$



Orientation O_y



Weight Map
 $W_{ls} = \hat{E}_y \odot B$

LSM Loss
(Magnitude Consistency)

LSO Loss
(Orientation Consistency)